

Q5-12_A Go to the five LearnChemE screencasts link for Chapter 5 (<http://www.umich.edu/~elements/6e/05chap/learn-cheme-videos.html>).

1. In the screencast of the PBR with pressure drop, is there a problem substituting for the volumetric flow rate when there is a change in the total number of moles, that is, $\delta \neq 0$?
2. View one of the other screencasts 5- to 6-minute video tutorials and list two of the most important points.
3. In the isothermal plug reactor screencast, how does the inert impact the reaction process?

① No there is not a problem where there is a change in total # of moles

② for an isothermal PFR:

1. Volumetric flow rate changes
2. Using the Ergun equation to account for the pressure drop.

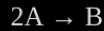
③ for gas phase Inerts doesn't effect our Concentration of A over the reaction because no moles of Inert are added.

for m

$$C_A = \frac{C_{A0} - \nu_A C_{A0} X}{1 + \epsilon_X} \left(\frac{F_{I0}}{F_T} \right) \left(\frac{T_0}{T} \right) \left(\frac{P}{P_0} \right)$$

$F_{I0} = F_T (z=z)$

P5-5A The elementary gas-phase reaction



is carried out in a constant-volume batch reactor where 50% conversion is achieved in 1 hour. Pure A is charged to the reactor at an initial concentration of 0.2 mol/dm^3 . If the same reaction is carried out in a CSTR, what volume would be necessary to achieve 50% conversion for a feed molar flow rate of 500 mol/h and an entering concentration of A of 0.2 mol/dm^3 ? (Ans: $V = 5000 \text{ dm}^3$)

P5-8B OEQ (Old Exam Question). The elementary gas-phase reaction



takes place isobarically and isothermally in a PFR where 63.2% conversion is achieved. The feed is pure A. It is proposed to put a CSTR of equal volume upstream of the PFR. Based on the entering molar flow rate of A to the first reactor, what will be the intermediate conversion from the CSTR, X_1 , and exit conversion from the PFR, X_2 , based on the feed to first reactor? The entering flow rates and all other variables remain the same as that for the single PFR. Source: Modified California Registration Exam Problem. (Ans: $X_2 = 0.82$)

BR:

$$\frac{N_A}{V_0} \frac{dx}{dt} = -r_A \frac{V_0}{V_0} \quad -r_A = k C_A^2$$

$$C_{A0} \frac{dx}{dt} = -r_A \quad C_A = C_{A0}(1-x)$$

$$\frac{dx}{dt} = \frac{-r_A}{C_{A0}}$$

$$\frac{dx}{dt} = \frac{k C_{A0}^2 (1-x)^2}{C_{A0}}$$

$$\frac{dx}{dt} = k C_{A0} (1-x)^2$$

$$\int_0^x \frac{dx}{(1-x)^2} = \int_0^t k C_{A0} dt$$

$$\left(\frac{x}{1-x} \right) = t k C_{A0} \quad \left. \begin{matrix} t = 1 \text{ hr} \\ x = 0.50 \\ C_{A0} = 0.2 \frac{\text{mol}}{\text{dm}^3} \end{matrix} \right\}$$

$$1 \text{ hr} = 0.20 K \left(\frac{1-0.50}{0.50} \right)$$

$$1 \text{ hr} = 0.20 \frac{\text{mol}}{\text{dm}^3} K$$

$$\frac{\text{dm}^3}{\text{mol}} = K$$

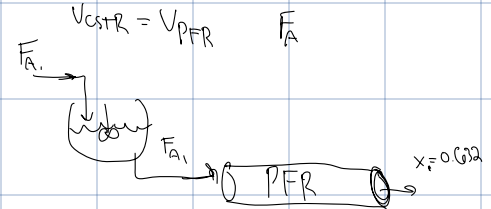
CSTR:

$$V = \frac{F_{A0} X}{-r_A(x)}$$

$$V = \frac{500 \frac{\text{mol}}{\text{hr}}}{k C_{A0}^2 (1-x)^2} X$$

$$V = \frac{500 \frac{\text{mol}}{\text{hr}}}{5(0.2)^2 (1-0.5)^2} (0.50)$$

$$V = 5000 \text{ dm}^3$$



PFR (Basis):

$$V_{PFR} = \int_0^{0.632} \frac{F_{A0}}{k C_{A0} (1-x)} dx = -\frac{F_{A0}}{k C_{A0}} \ln(1-0.632) = \frac{F_{A0}}{k C_{A0}} (0.9997)$$

$$V_{PFR} = V_{CSTR}$$

$$V_{CSTR} = \frac{F_{A1}}{k C_{A0} (1-x_1)} X_1$$

$$0.9997 \frac{F_{A0}}{k C_{A0}} = \frac{F_{A1}}{k C_{A0} (1-x_1)} X_1$$

$$0.9997(1-x_1) = X_1$$

$$0.4999 = X_1$$

PFR2

$$V_{PFR2} = \int_{X_1}^{X_2} \frac{F_{A0}}{-r_A(x)} dx = \int_{X_1}^{X_2} \frac{F_{A0}}{k C_{A0} (1-x)} dx$$

$$V_{PFR2} = \frac{F_{A0}}{k C_{A0}} \ln \left(\frac{1-X_2}{1-X_1} \right)$$

$$V_{CSTR} = V_{PFR2}$$

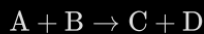
$$\frac{F_{A0}}{k C_{A0} (1-x_1)} X_1 = \frac{F_{A0}}{k C_{A0}} \ln(1-x_2)$$

$$0.9997 \frac{F_{A0}}{k C_{A0}} = \frac{F_{A0}}{k C_{A0}} \ln \left(\frac{1-x_2}{1-x_1} \right)$$

$$(1-x_2) = \frac{1-0.4999}{2.71}$$

$$x_2 = 0.816 \approx 0.82$$

P5-11_B OEQ (Old Exam Question). The irreversible elementary gas-phase reaction



is carried out isothermally at 305 K in a packed-bed reactor with 100 kg of catalyst.



The entering pressure was 20 atm and the exit pressure is 2 atm. The feed is equal molar in A and B and the flow is in the

turbulent flow regime, with $F_{A0} = 10$ mol/min and $C_{A0} = 0.4$ mol/dm³. Currently 80% conversion is achieved. What would be the conversion if the catalyst particle size were doubled and everything else remained the same? (Ans: $X = 0.83$)

$$C_A = \frac{(C_{A0} - C_{A0}X) \left(\frac{V_0}{V} \right)}{1 + 0X} = C_{A0}(1-X) \left(\frac{\frac{F_{A0}}{F_T} \left(\frac{T_0}{T} \right)}{\frac{F_{A0}}{F_T} \left(\frac{T_0}{T} \right)} \right) \left(\frac{P}{P_0} \right) = C_{A0}(1-X)P$$

$$C_B = \frac{(C_{B0} - C_{B0}X) \left(\frac{V_0}{V} \right)}{1 + 0X} = C_{A0}(1-X) \left(\frac{\frac{F_{A0}}{F_T} \left(\frac{T_0}{T} \right)}{\frac{F_{A0}}{F_T} \left(\frac{T_0}{T} \right)} \right) \left(\frac{P}{P_0} \right) = \frac{C_{A0}(1-X)P}{2}$$

$$-r'_A = k C_A C_B = k \left(\frac{C_{A0}(1-X)P}{2} \right)^2 \quad \frac{F_{A0}}{V_0} = C_{A0}$$

$$F_{A0} = C_{A0} V_0$$

$$F_{A0} \frac{dX}{dW} = k \left(\frac{C_{A0}(1-X)P}{2} \right)^2$$

$$\frac{dX}{dW} = \frac{k}{F_{A0}} \left(\frac{C_{A0}^2 (1-X)^2 P^2}{4} \right)$$

$$\phi = \frac{V_{\text{void}}}{V_{\text{solid}}} \quad 1 - \phi = \frac{V_{\text{solid}}}{V_{\text{void}}}$$

$$V_{\text{void}} = (1 - \phi) V_{\text{bed}} = (1 - \phi) \int_0^z \int_0^{2\pi} \int_0^R r dr d\theta dz$$

$$V_{\text{solid}} = (1 - \phi) \pi R^2 z$$

$$\frac{dP}{dz} = \frac{-G}{8gcD_p} \left(\frac{1 - \phi}{\phi^3} \right) \left[\frac{K_0(1 - \phi)\mu}{D_p} + 1.75G \right] \bigg|_{P=P_0}$$

$$\frac{dP}{dz} = - \frac{P_0}{P} \frac{1}{P} \left(\frac{F_T}{F_{A0}} \right) \left(\frac{T}{T_0} \right)$$

$$\beta_0 = \frac{G}{8gcD_p} \left(\frac{1 - \phi}{\phi^3} \right) \left[\frac{K_0(1 - \phi)\mu}{D_p} + 1.75G \right]$$

$$W = \underbrace{(1 - \phi) A_c z}_{\text{Volume of solid}} \cdot \underbrace{g_c}_{\text{Weight Catalyst}}$$

$$\frac{dP}{dz} \bigg|_z = \frac{W}{(1 - \phi) \pi R^2 z g_c} = \frac{dP}{dW} = - \frac{\alpha}{2P} \frac{T}{T_0} \frac{F_T}{F_{A0}}$$

$$\alpha = \frac{2\beta_0}{\pi R^2 g_c (1 - \phi) P_0} \quad \text{When } r = 2r \quad \alpha_1 = \frac{2\beta_0}{\pi r^2 (1 - \phi) P_0} = \frac{\beta_0}{2\pi r^2 (1 - \phi) P_0}$$

$$\text{if } e \rightarrow 0 \quad \frac{T}{T_0} = 1$$

$$\frac{\alpha_1}{\alpha} = \frac{1}{2}$$

$$\alpha_1 = \frac{1}{2} \alpha$$

$$\frac{dP}{dW} = - \frac{\alpha}{2P}$$

$$-\alpha = \frac{dP}{dW} \frac{2P}{P}$$

$$-\alpha = \frac{dP^2}{dW}$$

$$P^2 = (1 - \alpha W)$$

$$P = \sqrt{1 - \alpha W}$$

$$\frac{dX}{(1 - X)^2} = \frac{k C_{A0}}{V_0} P^2 dW$$

$$\int_0^{0.80} \frac{dX}{(1 - X)^2} = \frac{k C_{A0}^2}{F_{A0}} \int_0^{100} (1 - \alpha W) dW$$

$$\frac{1}{u^2} \quad u = 1 - X \quad du = -dX \quad -du = dX$$

$$-\int_{0.2}^{0.80} \frac{du}{u^2} = \frac{1}{u} \bigg|_{0.2}^{0.80} = \frac{1}{0.2} - \frac{1}{0.80} = \frac{1}{0.2} - \frac{1}{0.80} = \frac{1}{0.2} - \frac{1}{0.80}$$

$$\frac{1}{0.2} = \frac{k}{0.2} \left(\frac{0.4^2}{10} \right) \left(100 - \alpha \frac{100^2}{2} \right)$$

$$\frac{1}{0.2} \left(\frac{10}{0.4^2} \frac{1}{100 - 5000\alpha} \right) = k \quad \frac{312.5}{100 - 5000\alpha} = k$$

$$P = \frac{2}{20} = 0.1 = \sqrt{1 - \alpha(100)} \quad \alpha = 0.0049$$

$$\text{When } D_p = 2D_p, \alpha_1 = \frac{1}{2}\alpha$$

$$\int_0^X \frac{dX}{(1 - X)^2} = \frac{k C_{A0}^2}{F_{A0}} \int_0^{100} (1 - \alpha_1 W) dW \quad C_1 = \frac{C_{A0}^2}{F_{A0}} \left(\frac{312.5}{100 - 5000\alpha} \right) = \frac{5}{100 - 5000\alpha}$$

$$\frac{1}{1 - X} = C_1 \left(100 - \alpha_1 \frac{100^2}{2} \right)$$

$$\frac{1}{1 - X} = C_1 \left(100 - \alpha_1 \frac{100^2}{2} \right)$$

$$\frac{1}{1 - X} = \frac{5 \left(100 - \alpha_1 \frac{100^2}{2} \right)}{100 - \alpha \frac{100^2}{2}} \quad \alpha_1 = \frac{1}{2}\alpha$$

$$\frac{1}{1 - X} = \frac{5 \left(100 - \frac{\alpha}{2} \frac{100^2}{2} \right)}{100 - \alpha \frac{100^2}{2}} \quad \alpha = 0.0049$$

$$\frac{1}{1 - X} = 7.45$$

$$0.865 = X$$

P5-1B What if... you were asked to explore the example problems in this chapter to learn the effects of varying the different parameters? This sensitivity analysis can be carried out by downloading the examples from the CRE Web site (www.umich.edu/~elements/6e/index.html, under Living Example Problems (LEPs)). You can download **Wolfram** for free at <http://www.wolfram.com/cdf-player/>. For each of the example problems you investigate, write a paragraph describing your findings.

It is important to the learning process that you try to predict how the results change the answer or the shape of the conversion profile before moving a slider.

2. Example 5-4: Pressure Drop in a Packed Bed

Wolfram and Python

1. Vary each of the parameters in the Ergun Equation and write a couple of sentences on the variable that changes the slope of p versus z the most, that is, what change increases the pressure drop the most?
2. What generalizations can you draw from this example?

2.1.) A_c changed the slope of p vs. z the most followed by D_p , mass flow rate and μ which changed the slope the least.

2.2.) $\frac{p}{p_0}$ from inlet and outlet change dramatically when A_c or D_p are varied on the condition $\epsilon = 0$

3.1.) As K increases X also increases.

As v_0 increases $\left(\frac{p}{p_0}\right)_x = \left(\frac{p}{p_0}\right)_{x=0}$ but conversion decreases across W .

$$3.2.) \frac{dp}{dW} = -\frac{\alpha}{2p} \Rightarrow p = \sqrt{1 - \alpha W}$$

$$p = \sqrt{1 - (0.057)(25)} = 0.274$$

as $W \uparrow = \downarrow p$

$X \uparrow = \downarrow p \rightarrow$

3. Example 5-5: Effect of Pressure Drop on Conversion

Wolfram and Python

1. After varying each slider, write down a conclusion about the conversion and pressure profiles and explain why the slider movement gave you the results it did.
2. Why do C_A , C_B , and C_C all start to decrease at $W = 25$ kg?
3. What generalizations can you draw from this example?

5. Example 5-7: Calculating X in a Reactor with Pressure Drop

Wolfram and Python

This problem is a **Stop and Smell the Roses Simulation**. Explore this simulation to understand the interaction of pressure drop and conversion.



1. Explain what happens to X , p , and f as k' moves between its maximum and minimum values from its initial setting of $k' = 0.0035 \text{ s}^{-1}$.
2. What happens to X , p , and f as α moves between its maximum and minimum?
3. Write two conclusions from your slider experiments.
4. After reviewing *Generating Ideas and Solutions on the Web site* ([http://www.umich.edu/~elements/6e/toc/SCPS_3rdEdBook\(Ch07\).pdf](http://www.umich.edu/~elements/6e/toc/SCPS_3rdEdBook(Ch07).pdf)), choose one of the brainstorming techniques (e.g., lateral thinking) to suggest two questions that should be included in this problem.